

Nidhi A. Thakur

4th year Undergraduate | Life Sciences Major

 Email |  LinkedIn

EDUCATION

National Institute of Science Education and Research

Bhubaneswar

Integrated MSc. (3 years BSc. + 2 years MSc.) in Life Sciences

2023 - 2028

- 4-point GPA: 3.53/4
- SGPA (most recent semester): **9.1/10** (*ranked 1st in cohort*)

Secondary and Senior Secondary Education

Goa

Dempo - PACE; Goa Board of Higher Secondary Education (GBHSE)

2016 - 2023

- Grade 12 Marks: 95.0%

EXPERIENCE

Investigating role of Estrogen/Progesterone ratio on Temporal Lobe Epilepsy

Summer Intern

Dr. Sarat P Chandra and Dr. Jyotirmoy Banerjee, Epilepsy Neurobiology Lab, AIIMS Delhi June- July 2026

- Developed skills in processing of resected human brain tissue, Patch Clamp, Epilepsy model development in rats, ELISA, VEEG, and Histopathological assessment of samples

Field Electrophysiology for studying Synaptic Plasticity in Alzheimer's models

Winter Intern

Dr. Sreedharan, Synaptic Plasticity and Memory Lab, National University of Singapore Oct, Dec 2025

- Developed skills in recording and analysing field Excitatory Post-Synaptic Potentials (fEPSPs) in Rodent Hippocampal slices to explore plasticity impairments in Alzheimer's Disease models

Multimodal Assessment of Post Brain-Injury Alterations in Mice

Summer Intern

Dr. Kailash Manda, Defense Research and Development Organisation (DRDO), Delhi Jun - Jul 2025

- Selected to intern at INMAS's Neurobehavioural laboratory through the prestigious [IASc-INSA-NASI Summer Research Fellowship Program 2025](#), I investigated structural and neurobehavioural consequences of traumatic brain injury in rodent models using various neurobehavioural paradigms as well as histopathological techniques.

Regulation of Alternative Splicing in Hypothalamic GT1-7 cell-line using Polyamines

Intern

Dr. Arnab Banerjee, BITS Pilani - Goa

Jun-Jul 2024

- Studied the effect of the Polyamine Putrescine on the GnRH splice variant-1 and the RNA-binding protein NOVA 1 in the GT1-7 cell line, revealing an increased GnRH-V1 transcript expression without altering NOVA 1 expression when treated with Putrescine as opposed to the control. This research has potential significant applications in reproductive health, aging, and neurodegeneration.
- The internship was under the framework of the [BITS Goa Summer Research Program 2024](#) wherein I was one among the 26 candidates selected nationwide.

National Conference for Undergraduate Research in BioSciences (NCURB)

Poster Presenter

Indian Institute of Science, Education and Research (IISER) Tirupati

Dec 2024

- Selected among a competitive pool of national applicants to present a poster at IISER Tirupati based on my summer internship work at BITS Goa.

Ageing and The Brain: Impact of Neurodegenerative Diseases

Course Student

Universidad Complutense de Madrid

Apr-May 2025

- Gained a strong foundational understanding of the mechanisms driving aging and neuro-degenerative diseases, including Alzheimer's Disease and Parkinson's Disease.

FELLOWSHIPS

DST-INSPIRE Fellowship(Innovation in Science Pursuit for Inspired Research)	Awardee
Awarded to top 1% students in Grade 12 by Department of Science and Technology, Govt. of India Aug 2023 - 2028	
IASc-INSANA-NASI Summer Research Fellowship 2025	Awardee
Jointly awarded National-level Fellowship by National Science Academies: IASc-INSANA and NASI.	

ACHIEVEMENTS

National Entrance Screening Test (NEST)	All India Rank 185, Unreserved
NISER and UM-DAE CEBS	Jul 2023
Atal Tinkering Lab Marathon	Project Presenter
National Institution for Transforming India (NITI) Aayog	Sep 2018
– Presented our project ' <i>Smart Headlight</i> ' to Honorable PM Narendra Modi Ji	
Vidyarthi Vigyan Manthan	Goa State Topper
India's largest Digital Science Talent Search Examination organized by NCERT	2018
– Represented Goa at the national level held at Bhabha Atomic Research Centre (BARC), Trombay.	

EXTRACURRICULAR ACTIVITIES

The NeuroReach Initiative	Founder and Lead
Independent Outreach initiative to make brain science and health more accessible	2026 — Present
– Ongoing work includes collaborations with dementia care centres, educational media, and hospital outreach through awareness campaigns and canvas painting donations to wards. link to website	
Brain Matters — NISER's Neuroscience Club	President
National Institute of Science Education and Research (NISER)	May 2025 — Present
– Lead the planning and execution of journal clubs, alumni sessions, live demonstrations, and talks by professors from India and abroad to engage NISER's neuroscience community.	
Society for Neuroscience (SfN)	Member
	2025 — Present
Indian Academy of Neurosciences (IAN)	Member
	2026 — Present

SKILLS

In-Vivo:

Mice Handling, Neurobehavioural Testing

In-Vitro:

Preliminary skills in Mouse Brain Dissection and Sectioning, fEPSPs, Neuronal Cell Culture,

RNA-isolation, cDNA synthesis, Gel Electrophoresis, RT-PCR

Histopathological Techniques:

H and E staining, IHC, Golgi-Cox staining

Computational Techniques:

Python, NCBI-BLAST, Fiji (ImageJ)